

LinguaLink Play 2.0

Release and QA report

31 October 2026 · build 2.0.0 · for the LinguaLink360 team

Ready for a supervised pilot with real children

All 26 issues from the v1 report are fixed. 33 of 33 browser checks and 30 of 30 unit tests pass. Camera games take 19.4 s–26.9 s to start the first time (v1: 79–110 s) and 5.1 s–7.9 s for every level after that (v1: 56 s), including the short calibration, on a computer without a graphics card. Real-device testing and a native Urdu review remain before general clinical release (section 6).

26 / 26	33 / 33	30 / 30	5.1 s
v1 issues fixed	browser checks	unit tests	camera level start (v1 56 s)

From v1 to v2

Version 1 played correctly in touch mode but was not ready for children: camera games were very slow to start, leaving a loading game could save a fake session, Word Garden overscored unfinished levels, and the screens were written for therapists. Version 2 fixes every reported issue and rebuilds the experience around the child, with a separate area for adults.

Measured before and after

MEASURE	V1	V2
Fish Feast start, first time	79.5 s	**26.9 s**
Fish Feast start, next level	56.1 s	**5.1 s**
Firefly Night start, first time	110.2 s	**19.4 s**
Firefly Night start, next level	not measured	**7.9 s**
Downloads from outside servers per camera start	13–17 MB	**0 MB**
Pitch reading cost	18.3 ms	**about 0.4 ms**
First paint	1,976 ms	**1256 ms**
Phone home height	4,193 px	**2129 px**
Word Garden, 2 of 5 words	100 · 3 stars	**40 · 1 star**
Exit while a camera game loads	hidden game, result saved	**stops, nothing saved**

Same machine and method for both versions: Chrome 156 headless on Windows 11 with Chrome's fake camera and microphone. No graphics card was used, so real devices are expected to be similar or faster.

Your five notes on v1

Plain interface, no loading screen

Animated splash with a loading bar, a storybook home with five game tiles, level maps, stickers and rewards, a friendly loading card with progress for camera games, and a separate workspace for adults.

Slow loading

The bundled camera library was being rejected because of its file type, and the models came from Google every time. v2 ships the models, serves correct types, reuses the loaded model across levels, starts loading when the map opens, and caches everything for offline use.

Levels listed directly

Home shows games only. Each game has a five-stop map; the next level opens after finishing one with a star. Therapists can still pick any level from the Grown-ups area.

Repeated aaa / eee sounds

Balloon Breeze rotates four target sounds round by round. Sky Glide measures the voice range once per visit instead of every level. Each game has its own sound palette, and praise varies.

Report in JSON

The child sees stars and plain sentences. Adults get a session report with sentences, a chart, settings and difficulty changes, plus Save as PDF. JSON is kept for LinguaLink360 under Developer data.

2 · FIX REGISTER

All 26 v1 issues

17 are verified by automated tests that ship with v2 and can be re-run. The rest were checked by code review and are on the manual device checklist in TESTING.md.

ID	SEVERITY	V1 ISSUE	V2 CHANGE	VERIFIED BY	STATUS
C1	CRITICAL	Camera games took 56–110 s to start	Models and library ship with the app with correct file types; loaded once and reused; preload when the map opens; progress bar; offline cache	Browser test	Fixed
C2	CRITICAL	Exit during loading left a hidden game that saved a result	Each start-up step stops if the player has left; camera released; nothing saved	Browser test	Fixed
C3	CRITICAL	Word Garden scored 100 for 2 of 5 words	Score uses every item in the level; unfinished levels never suggest moving up	Browser + unit	Fixed
C4	CRITICAL	Pitch detection cost 18 ms per frame	Faster detector (about 0.4 ms), 20 readings a second, Sky Glide only	Unit test	Fixed
H1	HIGH	Word Garden could hear its own spoken word	Microphone ignores the game's own sounds and speech; echo cancellation on	Code review	Fixed
H2	HIGH	Result screen waited for the server	Result shows at once; 5 s send limit; outbox with retry	Browser test	Fixed
H3	HIGH	Delivery reported when it had not happened	Posts only to a known parent; waits for confirmation by result_id	Code review	Fixed
H4	HIGH	Sky Glide reported an invented pitch range	Asks again, otherwise reports Not measured	Unit test	Fixed
H5	HIGH	Urdu mode was mostly English	Every string and all game text translated; Urdu font included	Browser + unit	Fixed
M1	MEDIUM	A failed Breeze round was celebrated	Gentle sink and a try-the-next-one message	Code review	Fixed
M2	MEDIUM	Therapist mode timed out after 6–8 s	No time limit; Done, Not yet and Skip	Browser test	Fixed
M3	MEDIUM	Wrong /s/ letters highlighted	Exact target letters stored per word	Unit test	Fixed
M4	MEDIUM	Custom words ignored the level	Lists per level or per position	Unit test	Fixed
M5	MEDIUM	Privacy wording was inaccurate	Explains that the last try is kept until the next word	Code review	Fixed
M6	MEDIUM	All 25 levels listed at once	Five tiles, a level map per game, unlocking, therapist override	Browser test	Fixed
M7	MEDIUM	Not installable, no offline use	Icons, manifest and service worker	Browser test	Fixed

ID	SEVERITY	V1 ISSUE	V2 CHANGE	VERIFIED BY	STATUS
M8	MEDIUM	Waited for Google Fonts	Fonts hosted with the app and preloaded	Browser test	Fixed
M9	MEDIUM	Phone home page 4,193 px tall	Compact tiles; rotate prompt in portrait	Browser test	Fixed
L1	LOW	Countdown could not be paused	Pause in calibration, countdown and play	Browser test	Fixed
L2	LOW	Auto difficulty hid a result figure	Shown on its own line	Code review	Fixed
L3	LOW	Finger ring stayed after lifting	Cleared on lift	Browser test	Fixed
L4	LOW	Intro could name the wrong controls	Automatic mode mentions touch	Code review	Fixed
L5	LOW	Settings not keyboard friendly	Native dialogs; arrow keys on tabs	Code review	Fixed
L6	LOW	Reduced motion ignored in games	Still scenery, fewer particles	Code review	Fixed
L7	LOW	Unknown settings accepted	Validated with safe fallbacks	Unit test	Fixed
L8	LOW	Docs disagreed on test count	All docs rewritten for 2.0	Code review	Fixed

3 · BROWSER TESTS

33 of 33 checks pass

Run with `npm run test:e2e` on Sat, 31 Oct 2026 17:45:56 GMT.

AREA	CHECK	RESULT	DETAIL
M8	First paint without waiting for an outside font server	Pass	first paint 1256 ms, external requests: 0
UI	Splash screen with loading progress, then Tap to play	Pass	ready after 2101 ms
M6	Home shows 5 game tiles and no level lists	Pass	5 tiles, 0 level buttons on home
M9	Phone home fits the width and is short	Pass	scroll height 2129 px (v1: 4,193 px)
UI	Sticker book opens	Pass	
UI	Grown-ups area opens only after a 2-second hold	Pass	short tap opened: false, hold opened: true
M6	Level map: level 1 open, levels 2-5 locked	Pass	[false,true,true,true,true]
M6	Tapping a locked level explains how to open it	Pass	Finish level 3 to open this one
UI	Level intro card opens from the map	Pass	
REG	All 25 levels start, play and finish in touch mode with no script errors	Pass	25/25
H2	Result screen appears straight away	Pass	slowest 155 ms
C3	Word Garden: 2 of 5 words is not 100	Pass	L1: 40, easier; L2: 33, easier; L3: 33, easier; L4: 33, easier; L5: 33, easier
JSON	Result screen is plain language, with no JSON for the child	Pass	
REG	A level stopped early never suggests moving up	Pass	
L1	Pause works during the countdown	Pass	{"s":"countdown","p":true}

AREA	CHECK	RESULT	DETAIL
L3	Firefly touch cursor disappears when the finger lifts	Pass	
M2	Fish Feast therapist mode waits with no time limit and has Not yet	Pass	after 10 s: 0 moves timed out
JSON	Session report: plain sentences, chart, Save as PDF, JSON hidden	Pass	{"lines":4,"placeholders":false,"chart":true,"pdf":true,"jsonHidden":true}
H5	Urdu: game names, level text and buttons are in Urdu	Pass	English words left on the intro card: none
MIC	breeze: microphone mode starts	Pass	4718 ms to play, 56 fps
MIC	glide: microphone mode starts	Pass	4484 ms to play, 60 fps
MIC	garden: microphone mode starts	Pass	4567 ms to play, 60 fps
C1	Camera library and models load from this server, not CDNs	Pass	0 outside requests
C1	Camera game shows a progress bar on first load	Pass	
C1	Fish Feast starts in camera mode	Pass	first 26858 ms (v1: 79,473 ms)
C1	Second camera level starts fast (model reused)	Pass	5097 ms (v1: 56,092 ms)
C1	Firefly Night starts in camera mode	Pass	first 19411 ms (v1: 110,238 ms)
C1	Second Firefly level starts fast	Pass	7852 ms
C2	Leaving during loading: no hidden game, camera off, nothing saved	Pass	{"live":0,"results":0,"history":0,"gameHidden":true,"runner":false}
M7	Installable (icons 192 and 512) and works offline	Pass	service worker: true, offline home: true, icons: ["any","192x192","512x512","512x512"]

AREA	CHECK	RESULT	DETAIL
H2	Unreachable server: result shows at once and is queued for later	Pass	result shown in 171 ms; note: Saved on this device · Not sent yet. It will be sent when the connection is back.
SEC	Security headers (CSP, Permissions-Policy, nosniff)	Pass	
C1	Camera library is served as JavaScript	Pass	text/javascript; charset=utf-8

All 25 levels, and 30 unit tests

Each level was started in touch mode, played briefly and ended early on purpose, so low scores are expected. “Next level” is never suggested for a level stopped early.

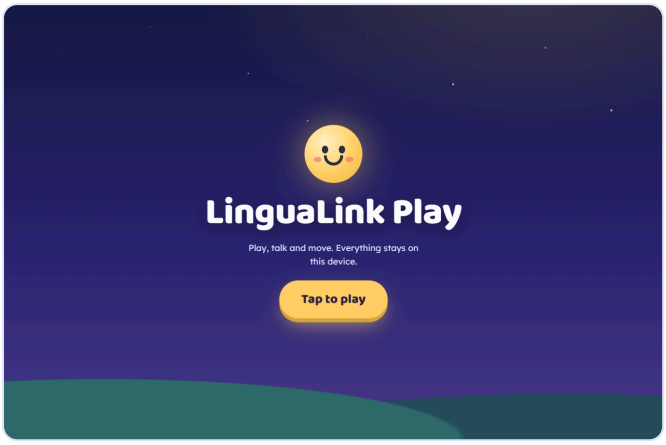
GAME	LEVEL	START TO PLAY	RESULT SHOWN	SCORE	SUGGESTION	SCRIPT ERRORS
breeze	1	2722 ms	91 ms	33	easier	0
breeze	2	2698 ms	62 ms	0	easier	0
breeze	3	2577 ms	29 ms	0	easier	0
breeze	4	2724 ms	99 ms	15	easier	0
breeze	5	2723 ms	96 ms	15	easier	0
glide	1	2663 ms	155 ms	0	easier	0
glide	2	2733 ms	18 ms	0	easier	0
glide	3	2723 ms	27 ms	0	easier	0
glide	4	2674 ms	9 ms	0	easier	0
glide	5	2724 ms	92 ms	0	easier	0
mouth	1	2688 ms	87 ms	40	easier	0
mouth	2	2696 ms	94 ms	33	easier	0
mouth	3	2676 ms	21 ms	25	easier	0
mouth	4	2749 ms	98 ms	25	easier	0
mouth	5	2730 ms	20 ms	20	easier	0
firefly	1	2684 ms	94 ms	0	easier	0
firefly	2	2707 ms	93 ms	0	easier	0
firefly	3	2727 ms	14 ms	0	easier	0
firefly	4	2751 ms	98 ms	0	easier	0
firefly	5	2682 ms	86 ms	0	easier	0
garden	1	2745 ms	59 ms	40	easier	0
garden	2	2708 ms	91 ms	33	easier	0
garden	3	2689 ms	91 ms	33	easier	0
garden	4	2700 ms	62 ms	33	easier	0
garden	5	2706 ms	112 ms	33	easier	0

Unit tests (30 of 30 pass)

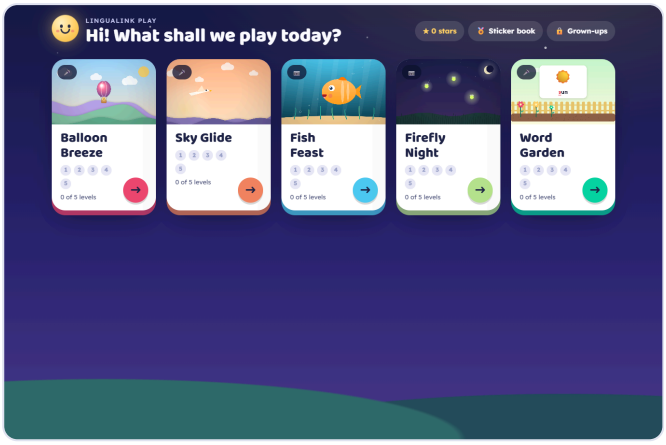
gets harder after a run of successes, easier after a run of misses	Pass
a miss resets the success streak and the other way round	Pass
steps stay inside the limits	Pass
disabled adaptive never changes	Pass
scale moves values in the right direction	Pass
no 'next level' suggestion if the level had to be made easier	Pass
Word Garden cue names follow the cueing hierarchy	Pass
Fish Feast threshold scale makes moves easier or harder to trigger	Pass
every game has 5 levels with a name, goal and settings	Pass
stars and suggestions follow the score bands	Pass
result object uses the shared schema	Pass
pitch detector finds a 220 Hz tone	Pass
loudness helpers	Pass
Balloon Breeze scoring	Pass
Sky Glide ring patterns and pitch mapping	Pass
Fish Feast move sequences and detection	Pass
Firefly Night zones and scoring	Pass
Word Garden word bank, highlighting and scoring	Pass
hand labels and blendshapes	Pass
C3: Word Garden scores against every item in the level	Pass
C4: pitch detection is accurate and fast	Pass
H4: Sky Glide never invents a pitch range	Pass
M3: target letters are marked exactly, no /sh/ or /z/	Pass
M4: custom words per level and per position	Pass
Balloon Breeze uses varied sounds round by round	Pass
Progress: levels unlock in order and stickers are awarded once	Pass
Reports: plain sentences for every game, no JSON needed	Pass
Results carry a unique id for delivery confirmation	Pass
L7: unknown settings fall back to safe values	Pass

H5: every English string and every game text has an Urdu version	Pass

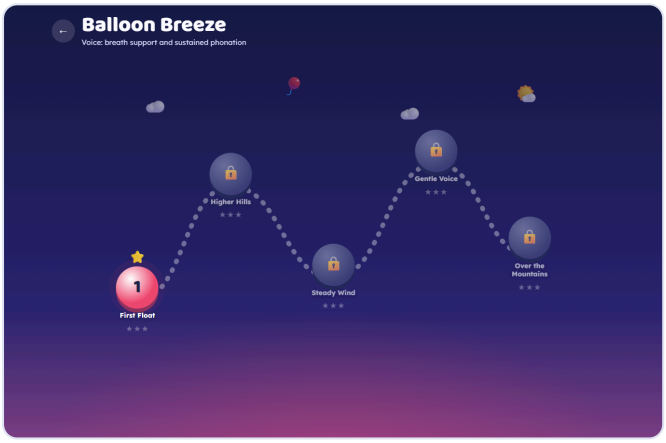
What v2 looks like



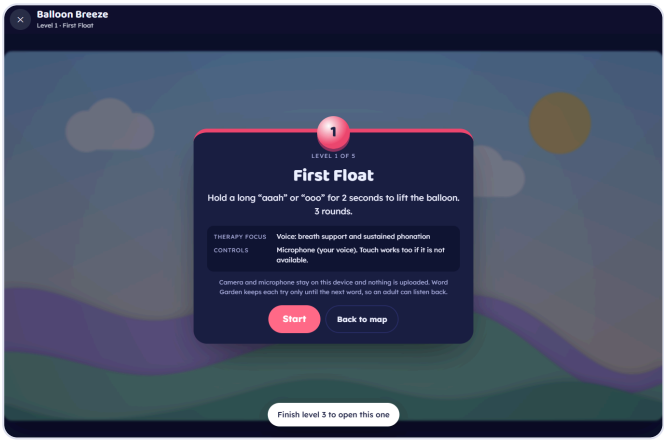
Splash screen with loading bar



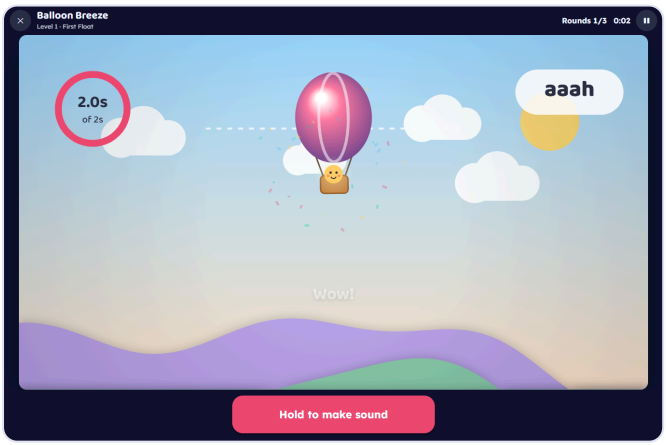
Home: five game tiles



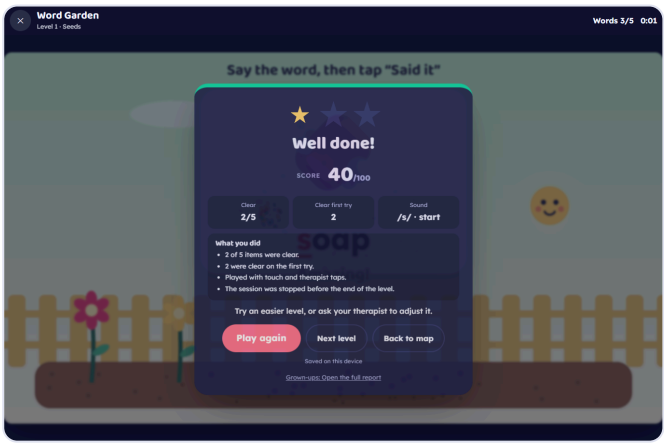
Level map with locked levels



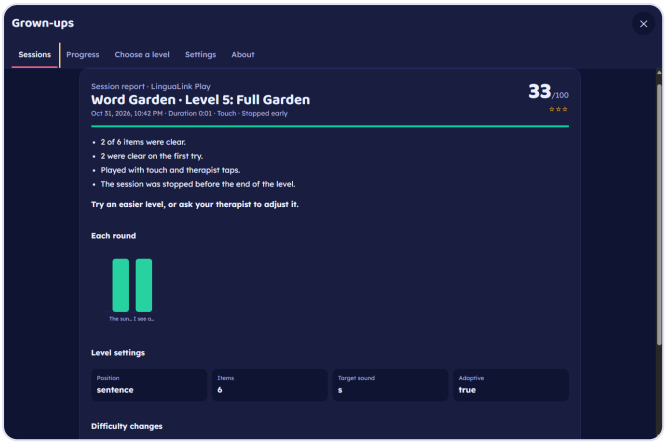
Level intro card



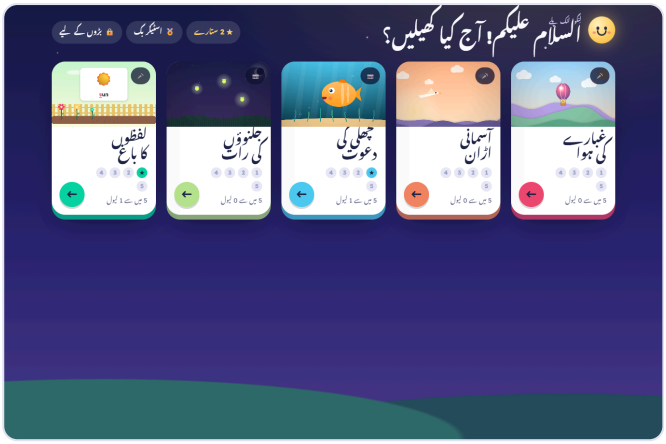
Balloon Breeze: the sound bubble changes each round



Result: plain sentences; 2 of 5 words scores 40



Grown-ups: session report



Urdu, right to left

6 · RELEASE

Shipping and next steps

Deploy

1. Push the `lingualink-play-v2` folder to a GitHub repository.
2. On Render: New > Blueprint, choose the repository. `render.yaml` creates an HTTPS Node service.
3. Check `/healthz` answers ok, then open the site on a tablet.
4. For LinguaLink360 embedding, set `FRAME_ANCESTORS` and add `parent=` to game links (DEPLOY.md).

Re-run the checks

- `npm test`: 30 unit tests.
- `npm start` then `npm run test:e2e`: 33 browser checks with screenshots.
- TESTING.md: manual checklist for real phones and tablets.

Known limits before general clinical use

- Tested with Chrome's fake camera and microphone. Face and hand thresholds, especially cheek puff, still need a pilot with real children on real phones and tablets.
- Urdu text was written for this release and needs review by a native speaker. Word Garden practises English words; Urdu lists can be added through customWords.
- Delivery confirmation to LinguaLink360 (H3) needs the small change in integration/LARAVEL.md on the LinguaLink360 side, then an end-to-end test there.
- On Render's free plan the service sleeps after 15 minutes idle; the first visit after that can take up to a minute. The offline cache hides this for returning devices; a paid instance avoids it.
- Scores are practice data that support the therapist's judgement. They are not a diagnosis.

Method

Code review of every source file; 30 unit tests with Node v24.14.1; 33 scripted checks in Chrome 156 headless with fake camera and microphone, served by the v2 Node server on Windows 11. v1 figures come from the v1 QA run on the same machine.